

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Second Semester B.Tech (CBCS) Degree Examinations

May 2024

MA1007-1 – DISCRETE MATHEMATICS AND TRANSFORM TECHNIQUES

(AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each **Unit**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- According to De Morgan's theorem, $\neg(p \wedge q)$ is logically equivalent to _____.
 A) $\neg p \wedge \neg q$ B) $\neg p \vee \neg q$
 C) $\neg p \Rightarrow \neg q$ D) $\neg p \equiv \neg q$
- In a valid form of disjunctive syllogism, if ' $p \vee q$ ' is the first premise and ' $\neg p$ ' is the second premise, the conclusion will be $\therefore$ _____.
 A) $\neg p$ B) $\neg q$
 C) $p \wedge q$ D) q
- $\neg(A \vee q) \wedge (A \wedge q)$ is a _____.
 A) Always True B) Always False
 C) q D) $\neg q$
- Which one of the following formulae in predicate calculus is valid?
 A) $\neg \forall x P(x) \equiv \forall x \neg P(x)$ B) $\neg \forall x P(x) \equiv \exists x P(x)$
 C) $\neg \exists x P(x) \equiv \exists x \neg P(x)$ D) $\neg \exists x P(x) \equiv \forall x \neg P(x)$
- Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 2)\}$ is a relation on A . Then R is :
 A) Reflexive B) Symmetric
 C) Transitive D) None of these
- An undirected graph contains a Euler's circuit if and only if
 A) All it's vertices are of even degree B) Exactly two of it's vertices are of even degree
 C) All it's vertices are of odd degree D) Exactly two of it's vertices are of odd degree
- A graph contains 16 edges, with 3 vertices of degree 4 and all other vertices of degree 2. The number of vertices of degree 2 are:
 A) 10 B) 4
 C) 2 D) 5
- The chromatic number of C_5 is
 A) 2 B) 3
 C) 4 D) 5
- The method which approximates the curve to the tangent to find root of an algebraic equation is:
 A) Newton-Raphson method B) Taylor's series method
 C) Regula-Falsi method D) Euler's method
- The initial approximation $y_1^{(0)}$ in the modified Euler's method is
 A) $y_0 + hf(x_0, y_0)$ B) $y_0 - hf(x_0, y_0)$
 C) $y_0 + f(x_0, y_0)$ D) $y_0 - f(x_0, y_0)$

11. Given the equation $x^3 + x - 3 = 0$, which of the intervals below contains a root?
 A) $[-1, 1]$ B) $[0, 1]$
 C) $[2, 3]$ D) $[1, 2]$
12. For the heat equation $u_t = \frac{1}{2} u_{xx}$, $h = k = 1$, the value of mesh ratio parameter α is :
 A) A) 1 B) $\frac{1}{2}$
 C) C) 0.25 D) 2
13. The Fourier series expansion of an odd periodic function contains
 A) Only cosine terms B) Cosine terms and a constant
 C) Only sine terms D) Sine terms and a constant
14. The value of a_0 in the half range Fourier cosine series expansion of $f(x) = 2x$ in $(0, 3)$ is
 A) 0 B) 3
 C) 6 D) 9
15. If $F(x(t)) = X(\omega)$, is the Fourier transform of the function $x(t)$, then $F(x(at)) =$
 A) $\frac{1}{a} X\left(\frac{\omega}{a}\right)$ B) $aX(a\omega)$
 C) $\frac{1}{a} X(a\omega)$ D) $aX\left(\frac{\omega}{a}\right)$
16. Which of the following function is odd
 A) $f(x) = \cos x$; $-\pi < x < \pi$ B) $f(x) = x^2$; $-\pi < x < \pi$
 C) $f(x) = k$; $-\pi < x < \pi$ D) $f(x) = x$; $-\pi < x < \pi$
17. If a, b, c are constants and u_n, v_n, w_n are any discrete functions, then $Z(au_n + bv_n - cw_n)$ is:
 A) $aZ(u_n) + bZ(v_n) - cZ(w_n)$ B) $aZ(u_n) + bZ(v_n) - cZ(w_n)$
 C) $aZ(u_n) - bZ(v_n) - cZ(w_n)$ D) $aZ(u_n) + bZ(v_n) + cZ(w_n)$
18. If u_n is a function of n then the unilateral Z-transform of u_n is defined as, $Z(u_n) =$
 A) $\sum_{n=0}^{\infty} u_n z^{-n}$ B) $\sum_{n=0}^{\infty} u_n z^n$
 C) $\sum_{n=-\infty}^{\infty} u_n z^{-n}$ D) $\sum_{n=-\infty}^{\infty} u_n z^n$
19. $Z^{-1}\left(\frac{1}{z-2}\right)$ is :
 A) 2^n B) 2^{n-1}
 C) 2 D) $\frac{1}{2}$
20. If $Z(u_n) = U(z)$ then $Z(a^n u_n)$ is :
 A) $U\left(\frac{a}{z}\right)$ B) $U(za)$
 C) $U\left(\frac{z}{a}\right)$ D) $U(a^n z)$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

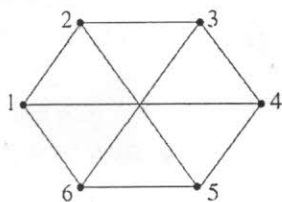
Unit – I		Marks	BT*	CO*	PO*
1. a)	Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.	6	L*2	1	2
b)	Consider the premises: "It's not sunny and it's colder than yesterday". "We will go swimming only if it's sunny." "If we don't go swimming then we will take canoe trip." "If we take a canoe trip, then we will be home by sunset." Prove that the conclusion: "We will be home by sunset." is valid.	6	L3	1	1
c)	Determine the truth value of each of these statements if the domain consists of all real numbers: i) $\exists x(x^3 = -1)$ ii) $\exists x(x^4 < x^2)$ iii) $\forall x((-x)^2 = x^2)$ iv) $\forall x(2x > x)$.	4	L1	1	1

2. a) Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg(p \vee q)$ are logically equivalent by developing a series of logical equivalences. 6 L3 1 1
- b) Use mathematical induction to show that:
 $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n . 6 L2 1 2
- c) Using a truth table, show that the compound proposition is a tautology:
 $\neg(p \rightarrow q) \Leftrightarrow p \wedge \neg q$. 4 L1 1 1

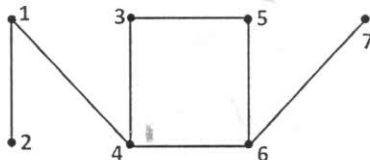
Unit – II

3. a) Let $A = \{1, 2, 3, 4, 6, 8\}$, R be a relation on A defined by aRb if and only if a divides b .
 (i) Find the relation R
 (ii) Draw the digraph of R
 (iii) Find the indegree and outdegree of each vertex
 (iv) Find the matrix M_R . 6 L2 2 1
- b) Define bipartite graphs and complete bipartite graphs. Check whether the following graphs are bipartite or not. If the graph is bipartite, give the bipartitions of each graph.

i)

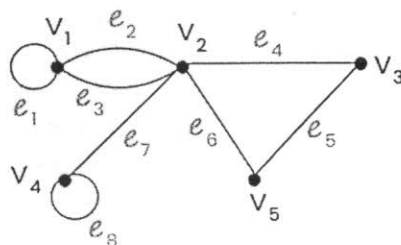


ii)



- c) Let $A = \{1, 2, 3, 4, 5, 6\}$ and $p_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 2 & 6 & 5 \end{pmatrix}$, $p_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 3 & 1 & 2 & 5 & 6 \end{pmatrix}$
 and $p_3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 3 & 1 & 6 & 5 \end{pmatrix}$. Compute (i) $(p_3 \circ p_2) \circ p_1$ (ii) $p_1^{-1} \circ p_2$ 4 L1 2 1

4. a) Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 3), (3, 1), (4, 4)\}$. Find the transitive closure of R using Warshall's algorithm. 6 L1 2 2
- b) What are planar graphs? Are the graphs K_4 and K_5 planar? Justify your answer with suitable diagrams. 6 L2 2 1
- c) Find the adjacency matrix and incidence matrix of the graph:

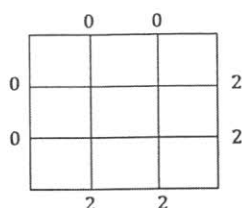


4 L2 2 1

Unit – III

5. a) Using Runge-Kutta method, find $y(0.1)$ given that $\frac{dy}{dx} = 3x + y$, $y(0) = 1$ and $h = 0.1$. 6 L1 3 1

- b) Solve $u_{xx} + u_{yy} = 0$ for the square mesh with the boundary values as shown below:



- c) Using the Newton Raphson method, find a real root of $x^2 - \sin x - 2$ taking an initial approximate $x_0 = 2$. Carry out 3 iterations.
6. a) Find the root of the equation $xe^x = \cos x$ using Regula-Falsi method correct to four decimal places in $(0,1)$. Carry out 3 iterations.
- b) Find the solution of the parabolic equation $u_t = \frac{1}{2}u_{xx}$ when $u(0,t) = u(4,t) = 0$ and $u(x,0) = x(4-x)$ taking $h = k = 1$. Find the values of u up to $t = 5$.
- c) Find an approximate value of y when $x = 0.1$, if:
- $$\frac{dy}{dx} = x + y^2, y(0) = 1 \text{ \& } h = 0.1$$
- using Taylor's series method up to third degree terms.

6 L2 3 2

4 L1 3 2

6 L2 3 2

6 L2 3 1

4 L1 3 1

Unit – IV

7. a) If $f(x + 2\pi) = f(x)$, find the Fourier series expansion of $f(x) = x + 1$ in the interval $(-\pi, \pi)$.
- b) Find the Fourier series expansion of $f(x) = |x|$; $f(x + 2\pi) = f(x)$.
- c) Obtain half range Fourier Cosine series for $f(x) = x^2$ in the interval $(0,1)$.
8. a) If $f(t)$ and $g(t)$ are piecewise continuous integrable functions the prove that $F\{(f * g)(t)\} = F\{f(t)\} \cdot F\{g(t)\}$.
- b) Find the Fourier sine transform of $f(t) = \frac{e^{-at}}{t}$, $a > 0$.
- c) Find the Fourier transform of $f(t) = \begin{cases} 1; & |t| \leq 1 \\ 0; & |t| > 1 \end{cases}$.

6 L2 4 1

6 L1 4 2

4 L2 4 1

6 L1 4 2

6 L2 4 1

4 L2 4 2

Unit – V

9. a) Find the Z – transform and ROC of the function $u_n = \begin{cases} 2^{-n}, & \text{if } n \geq 0 \\ 2^n, & \text{if } n < 0 \end{cases}$
- b) State and prove the convolution theorem for bilateral Z – transforms.
- c) Using the method of partial fractions, find the inverse Z – transform of $U(z) = \frac{z}{(z-2)(z-4)}$ in the region $2 < |z| < 4$.
10. a) State and prove the final value theorem for unilateral Z- transforms.
- b) Using Z – transforms, solve the difference equation:
 $u_{n+2} - 2u_{n+1} + u_n = 2^n$ with $u_0 = 2$ and $u_1 = 1$.
- c) State the initial value theorem for unilateral Z- transforms. Using this find the value of u_0 when $U(z) = \frac{z^2}{(2z+1)(z+1)}$.

6 L2 5 1

6 L1 5 2

4 L2 5 1

6 L1 5 2

6 L2 5 1

4 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
